

Share-k — AI-Powered Open Source Collaboration Hub

1. MVP Scope for the ITI Proposal

Project Name

Share-k — AI-Powered Open Source Collaboration Hub

Project Vision

Share-k is a platform that connects open-source project owners with developers who want to contribute, learn, and build their technical reputation, using AI-based skill analysis and contributor matching.

The platform enables developers to:

- discover real projects
- contribute to open source
- improve their technical skills
- build a verified contribution profile

At the same time, project owners can:

- publish their projects
- request contributions
- find suitable contributors using AI analysis.

The system also acts as a learning hub, allowing developers to identify the skills they need in order to contribute effectively to real projects.

2. User Roles

The MVP will include three main roles.

1. Project Owners

Project owners are developers who want others to contribute to their open-source projects.

They can:

- add GitHub projects
- create contribution tasks
- review contributor applications
- confirm completed contributions
- rate contributors

2. Contributors

Contributors are developers who want to gain experience through real projects, contribute to open source, improve their technical skills, build a verified reputation, and apply for paid contribution tasks.

Before an application is sent to the project owner, the system performs a basic AI skill validation to ensure that the contributor has the required technical background.

3. Admin / Support

Admins are responsible for:

- verifying new users
- validating AI-generated skills
- reviewing reported issues
- preventing fraud or misuse of the platform.

3. Core Features (MVP)

3.1 User Registration & Profile Creation

Users create an account and connect their GitHub profile.

The system automatically fetches:

- repositories
- programming languages
- contribution activity
- technologies used in projects.

The AI then generates a skill profile for the user.

Python – Advanced
React – Intermediate
Docker – Beginner

Admins review and approve the generated skills before the account becomes fully active.

3.2 Project Publishing

Project owners can add a project by providing:

GitHub Repository URL

The system automatically fetches repository information such as:

- project title
- description
- programming languages
- tags and technologies
- repository statistics.

The owner can review and edit this information before publishing.

After confirmation, the project becomes visible on the platform projects page.

3.3 Project Discovery

Contributors can browse all available projects on the platform.

They can filter projects by:

Technology
Category
Difficulty level

Example categories:

- Web Development
- Mobile Development
- AI / Machine Learning
- DevOps
- Tools & Utilities

3.4 Contribution Requests (Orders)

Project owners can create contribution tasks describing specific work needed in their project.

Each task includes:

Title
Description
Required technologies
Difficulty level
Deadline
Optional reward

Example:

Add authentication using JWT
Difficulty: Intermediate
Technologies: Node.js, JWT
Deadline: 5 days
Reward: \$80

Tasks appear on the orders page, where contributors can apply.

3.5 Applying for Contribution Requests

Contributors can apply to tasks they are interested in.

When a contributor applies, the system performs a simple AI skill validation to compare the contributor's skills with the task requirements.

Possible outcomes:

Eligible contributor

- If the contributor has sufficient skills
- the request is forwarded to the project owner
- the owner reviews the application and decides whether to accept it.

Not eligible (Regular user)

- If the contributor lacks the required skills
- the system informs the user that they are currently not eligible to contribute to this project.

This prevents project owners from receiving unsuitable applications.

3.6 Contribution Delivery

Once accepted, the contributor works on the task and submits a GitHub Pull Request link.

The owner can:

- approve the contribution
- reject the contribution
- rate the contributor
- leave feedback.

3.7 Reputation System

Each contributor builds a reputation profile based on their activity.

Metrics include:

completed contributions
ratings
success rate
skills

Example contributor profile:

Rating: 4.8
Tasks completed: 18
Success rate: 94%
Top skills: React, Node.js

A higher reputation increases the chances of being selected for future contributions.

4. Premium Packages (Included in MVP)

The platform provides premium subscriptions for both project owners and contributors, unlocking advanced features and increased usage limits.

4.1 Owner Premium Plans

Bronze Plan

Basic premium plan with limited advantages.

Up to 10 contribution orders per month
Standard project visibility

Silver Plan

Mid-level plan with improved exposure.

Up to 20 contribution orders per month
AI contributor matching (Top 5)
Priority visibility in project listings

Gold Plan

Highest premium plan for project owners.

Up to 30 contribution orders per month
AI contributor matching (Top 10)
Priority order visibility
Automatic notification to best-matching contributors
No platform commission on contributor payments

4.2 Contributor Premium Plans

Bronze Plan

Up to 2 applications per day
Basic task notifications

Silver Plan

Up to 3 applications per day
Skill-matched task notifications
Reduced platform commission

Gold Plan

Up to 4 applications per day
AI recommended tasks based on skills and activity
Skill gap feedback when rejected by AI validation
No platform commission
Priority application visibility

If a Gold contributor applies to a task but does not yet meet the required skills, the system provides AI guidance showing which skills are missing, recommended technologies to learn, and suggested projects to practice on.

This helps the contributor improve and become eligible for similar tasks in the future.